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Blockchain-Based Solutions for Enhancing Transparency and Efficiency in Organ Transplantation

Arvind Mewada, Sumit Kumar Sharma, Mohd. Aquib Ansari, Roshni Yadav, and Shahnawaz Ahmad

Abstract

Organ transplantation faces major challenges, including organ shortages, inefficient allocation, and ethical concerns. This study proposes a blockchain-based framework to enhance transparency, security, and efficiency in transplantation workflows. By leveraging decentralised ledgers, smart contracts, and cryptographic techniques, the system addresses data tampering, traceability gaps, and allocation delays. The multi-layered architecture integrates immutable records, automated matching, role-based interfaces, and IoT-enabled organ tracking. Simulation results demonstrate a 75% reduction in matching time compared to a conventional manual allocation system, as well as a 92.5% decrease in errors. The framework also supports ethical compliance and auditability and reduces illicit organ trade through verifiable custody chains. While promising, the approach faces challenges related to scalability, Electronic Health Records (EHR) interoperability, and regulatory harmonisation. Future directions include pilot implementations and Layer-2 enhancements for broader deployment.

Introduction

Organ transplantation relieves ailments and enhances the quality of life for many people. The need for transplantable organs continues to outpace the available supply despite advancements in surgical care and postoperative management. This lack of availability results in prolonged waiting times and a large number of potentially preventable deaths each year. This gap is a significant public health issue and highlights the lack of effective systems for managing the procurement, allocation, and transplantation of organs. An enhanced technique for organ procurement is therefore highly needed, as global demand continues to increase until alternative technologies for organ replacement are fully developed. While patients require organs across the globe, the availability rate is extremely scarce, leading to a humanitarian catastrophe [1]. With the limited number of available organs, the selection of transplant recipients is particularly challenging due to factors such as age, lifestyle, and medical urgency. These challenges are further compounded by ethical concerns, which include obtaining proper consent for organ donation, especially in posthumous cases. Organ trafficking introduces additional ethical dilemmas, such as whether donors should be compensated, which further complicates the transplantation process. Beyond surgical precision, several non-clinical factors affect the transplant ecosystem, including organ matching and proper preservation during transportation. Existing organ transplantation frameworks continue to struggle with ineffective approaches to addressing challenges such as transplant complications and inefficient allocation systems that impede progress [2].

Current organ transplantation systems primarily rely on centralised databases and manual workflows, which suffer from several weaknesses. This system is vulnerable to unauthenticated access, data breaches, cyberattacks, corruption, and a lack of transparency. These systems achieve improved transparency, security, and efficiency through the adoption of blockchain technology. The Blockchain is a decentralised ledger system that functions as an immutable database and provides solutions to many of the issues faced by current frameworks [3]. The Blockchain technology ensures transparent data integrity, thereby significantly increasing stakeholder trust. A data integrity framework that provides complete protection delivers both trust assurance and data assurance to stakeholders. As a result, blockchain technology demonstrates strong potential for the healthcare sector through transparent and traceable data interactions.

The Blockchain technology also assists in tracking organ donation information, thereby supporting efforts to combat organ trafficking. Health records of the Patients can be stored securely, enabling authorised clinical access while safeguarding confidentiality. In pharmaceutical supply chains, enhanced transparency through blockchain helps prevent the distribution of counterfeit drugs [4]. Additionally, blockchain enhances the management of clinical and drug development-related documentation by facilitating the sharing of inter-institutional information [5]. The

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methods proposed in this study aim to address patient needs while supporting effective healthcare delivery. This research focuses on resolving ethical, organisational, and logistical challenges in organ transplantation systems using blockchain technology. It automates donor–recipient matching through smart contracts, ensuring the secure processing of sensitive information, comprehensive organ traceability, and robust data privacy mechanisms. Smart contracts further reduce administrative costs and help curb unlawful organ trafficking through verifiable audit trails. The proposed framework encompasses the entire transplant pipeline, from donor registration to organ delivery, by integrating IoT technologies with dynamic queuing and self-executing agreements. Compared to existing approaches, the proposed system demonstrates improved performance, achieving a 75% increase in matching speed and a 92.5% reduction in errors. The main contributions of this paper include analysing failure modes in transplant systems, evaluating blockchain's strengths and limitations in healthcare data governance, proposing a transparent and efficient blockchain-based transplantation framework, examining ethical and non-technical adoption challenges, and providing guidelines to support pilot deployment and scalable implementation.

RELATED WORK

Organ transplantation faces major challenges, including global organ shortages, inefficient allocation methods, ethical concerns, and a lack of transparency. Regional disparities in organ availability result in long wait times and high mortality rates [4]. Traditional systems rely on outdated manual workflows and centralised databases, leading to delays, data breaches, and single points of failure [5]. Ethical issues, such as informed consent, fair access, and risks of exploitation in post-mortem cases, remain critical. Economic coercion further raises concerns about commodification and organ trafficking. The absence of real-time traceability and auditability allows room for fraud and erodes trust [2]. Blockchain offers a decentralised, immutable, and transparent framework, enabling secure, auditable, and tamper-proof records. Its integration improves data integrity and stakeholder confidence in organ allocation [4].

Blockchain for Data Security and Transparency

Daniel et al. [2] proposed a blockchain that incorporates a distributed file system, enabling decentralised waitlist management along with controlled access for medical record sharing, to enhance transparency while limiting information tampering and unauthorised access. Bharti et al. [5] identified ethical and security risks associated with organ donation, which were addressed using a blockchain-based framework that focuses on transparency, encryption, and immutability. Kypu et al. [6] present a decentralised application on Ethereum for donor registration and pledge validation via smart contracts. Bawa et al. [7] highlight the limitations of transparency in centralised systems, recommending Ethereum-based solutions for real-time access and auditability. Anselmo et al. [8] explore the use of NFTs for authenticating organs and ensuring traceability, suggesting innovative audit mechanisms to mitigate black-market activities. Among the platforms discussed, Hyperledger Fabric, with its permissioned and privacy-preserving design, is better suited for organ transplantation than public Ethereum, which may face limitations in terms of privacy and gas costs in clinical contexts.

Decentralised Organ Matching and Allocation Systems

Patchipulusu et al. [9] created an automated organ allocation and validation system using distributed ledgers and smart contracts. Igboanusi et al. [10] developed BOMS, an information-sharing system based on biological compatibility cross-matching, which is secured by Ethereum, demonstrating strong performance in cross-matching. Panda et al. [11] propose KidneyChain, a blockchain and AI-integrated platform that leverages clinician metrics and convolutional neural networks (CNNs) for organ suitability prediction and cybersecurity. Kumar and Jeysankar [12] combine AI and blockchain technologies to improve donor–recipient match accuracy and reduce wait times, presenting a Hyperledger Fabric-based prototype (OPTN) that evaluates chaincode schemes for efficient on-chain organ matching. While public Ethereum networks enable smart contract automation, Hyperledger Fabric is more appropriate for organ transplantation scenarios due to its support for granular access control, deterministic execution, and regulatory compliance, all of which are critical for healthcare governance.

Smart Contracts and Consent Verification

Jadhav and Pise [13] present an ABAC-integrated blockchain model ensuring secure donor access control and transparency. They also propose Smart Donations, an event-driven system utilising smart contracts for conditional and transparent donations. Sitharamulu et al. [14] developed a private Ethereum blockchain framework with six specialised smart contract algorithms for secure and private organ transfer. Additionally, they introduce a Hyperledger Fabric-based solution for tracking, consent verification, and interoperable EHR sharing to support cross-institutional trust. Consent verification on public Ethereum offers transparency but risks data exposure, whereas private blockchains, such as Hyperledger Fabric, ensure the secure, confidential, and verifiable execution of consent protocols.

Adoption Factors and Ethical Implications

Rajan et al. [4] demonstrate a decentralised web application using IPFS for secure medical data storage and blockchain to eliminate intermediaries. Panigrahi et al. [15] apply structural equation modelling to identify hospital adoption drivers, indicating blockchain's role in improving traceability and stakeholder trust. These studies also vary in blockchain architecture, with IPFS and Ethereum enabling decentralised storage and access. However, their integration into permissioned networks (e.g., Hyperledger) is essential for ensuring data control in sensitive healthcare domains.

Summary and Research Gaps

Together, these studies highlight blockchain's promising role in revolutionising organ transplantation by enhancing decentralisation, automation, and data integrity. Most works remain as proofs of concept or theoretical designs, with no practical, full-scale implementation being undertaken.

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There is insufficient focus on addressing ethical, logistical, and regulatory issues, and specifically, a lack of standardised cross-compatibility with other healthcare systems, seamless integration into EHRs, and comprehensive, holistic approaches. As summarised in Table 1, existing solutions are often fragmented, focusing on isolated components rather than end-to-end transplant workflows. Further development should focus on model validation through field testing and deployment at an institutional or national scale.

REF.	Platform	Feature	Limitation
[2]	Ethereum	Decentralised waitlist and record sharing	Privacy risks; no tracking or prioritisation
[5]	Ethereum	Secure donor data, transparency	Gas cost; static allocation
[6]	Ethereum	Donor registration via smart contracts	Limited scalability; no EHR/IoT
[8]	Blockchain + NFT	Organ authenticity and traceability	No clinical prioritisation
[9]	Blockchain	Automated allocation framework	Lacks a dynamic queue model
[10]	Ethereum	Compatibility-based organ matching	Public-chain privacy issues
[11]	Blockchain + AI	AI-assisted suitability prediction	Limited auditability
[12]	Hyperledger Fabric	Permissioned AI-based matching	No logistics/transport tracking
[14]	Private Ethereum	Smart-contract consent & EHR sharing	Limited scalability analysis
Our Work	Hyperledger Fabric	End-to-end allocation, priority queuing, IoT tracking, and auditability	Infrastructure & regulatory challenges

TABLE 1. Comparison of Blockchain-Based Organ Transplantation Solutions.

Proposed Model for Blockchain-Based Organ Transplantation

The review in the "Related Work" section indicates that most existing blockchain-based organ transplantation solutions address isolated components of the workflow, such as donor registration or matching, without providing an integrated operational pipeline. To overcome this limitation, we propose an end-to-end framework built on a permissioned blockchain that unifies donor–recipient matching, dynamic priority queuing, real-time organ tracking, and regulatory-compliant auditability. The proposed model consolidates technical and operational processes into a single ethically governed system, ensuring transparency, fairness, and accountability across the entire transplantation lifecycle.

Design Rationale and Core Objectives

The proposed framework is designed to address key functional and governance gaps in existing approaches, including static allocation mechanisms, limited traceability, and insufficient regulatory support. It provides a unified transplant workflow that integrates donor registration, recipient matching, allocation, organ transport, and implantation within a single system.

At its core, the framework enables fairness-driven allocation through a dynamic, multi-tier, real-time queuing mechanism that prioritises recipients based on clinically grounded criteria such as medical urgency, biological compatibility, and waiting time. Trust and accountability are ensured through tamper-proof traceability and cryptographically verifiable audit trails for each organ, enabling effective regulatory oversight and deterring illicit practices. Data governance is enforced using a permissioned blockchain architecture with Self-Sovereign Identity (SSI), fine-grained access control, and cryptographic protections aligned with healthcare regulations (e.g., GDPR and HIPAA). Additionally, interoperability with IoT-enabled transport sensors and hospital Electronic Health Record (EHR) systems enables real-time monitoring of organ viability and informed clinical decision-making.

Mathematical Foundations

The proposed framework formalises the organ allocation process using two fundamental functional components: donor-recipient compatibility matching and priority-based recipient selection, to support automated and transparent allocation decisions. These mathematical models provide the analytical basis for executing allocation decisions through smart contracts.

1) Matching Function F_{match} (d, r): The matching function determines the compatibility between a donor d and a recipient r. The compatibility is evaluated using established medical criteria, which include blood group compatibility (ABO and Rh factor alignment), human leukocyte antigen (HLA) tissue matching with a minimum histocompatibility threshold of 80% and medical urgency indicators (e.g., MELD score). Based on these criteria, the function produces a binary decision output, as defined in Eq. (1), which ensures that only clinically compatible donor–recipient pairs proceed to the prioritisation stage.

$$F_{match}(d,r)=\begin{cases}1, & \text{IF COMPATIBILITY_SCORE}(d,r)\geq \mathrm{T}\\ 0, & \text{otherwise,}\end{cases} \quad (1)$$

where τ represents the predefined compatibility threshold (set to 0.80 in this study).

2) Priority Function $G_{priority}$(r): Once compatibility is established, the priority function assigns a score to each eligible recipient. This score is computed using a weighted combination of clinically relevant factors. These factors are medical urgency ($\alpha = 0.5$), quantifying the severity of the patient's condition (e.g., MELD score, ICU status), wait time ($\beta = 0.3$), accounting for the duration the recipient has spent on the transplant list, and age factor ($\gamma = 0.2$), with higher preference assigned to paediatric patients. The resulting priority score is normalised to ensure equitable and consistent ranking across all candidates, as expressed in Eq. (2).

$$G_{\text{priority}}(r) = \alpha\cdot\text{urgency}(r)+\beta\cdot\text{wait time}(r)+\gamma\cdot\text{age}(r) \quad (2)$$

This score is then normalised within the range of 0 to 100 to ensure equitable and consistent prioritisation across all candidates.

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System Architecture

The proposed blockchain-based organ transplantation system adopts a layered architecture, as illustrated in Fig. 1, to operationalise the design objectives and mathematical models described above. The architecture integrates data security, automated decision logic, application-level interaction, and real-time sensing to support secure and decentralised organ allocation.

- Data Layer: The security architecture of this layer utilises Hyperledger Fabric technology to establish permissioned access for storing non-modifiable donor, recipient, and organ metadata records. The technology utilises SHA-256 hashing to maintain data integrity and employs Self-Sovereign Identity (SSI) for private digital identity management.
- Smart Contract Layer: This layer executes the matching function F_{match} (d, r) and the priority function $G_{\text{priority}}(r)$ to automatically perform rule-based organ allocation.
- Application Layer: Dashboard interfaces tailored for roles exist to benefit hospitals and regulatory groups. The system enables real-time tracking of organ availability while controlling user access through roles to present stakeholders, including surgeons' coordinators and auditors, with only their relevant information.
- IoT Integration: The integration of GPS technology with biosensors within organ transport units helps improve trackability. Real-time environmental data, including temperature, humidity, and location information, is captured and transmitted through devices that store the data on-chain to ensure transport integrity and regulatory compliance.

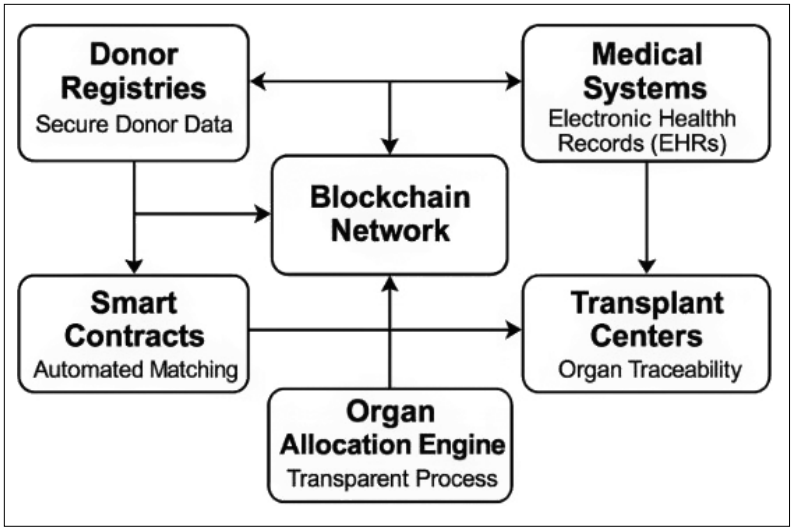


FIGURE 1. Blockchain-Based Organ Transplantation Workflow.

Input: $d \in D$, $r_i \in R$, and $o \in O$
Output: Allocated recipient r^*

Authenticate donor and organ data by verifying digital signatures σd and σo using cryptographic functions and one-time passwords (OTP).
Initialise an empty eligible recipient set $R_{eligible} \leftarrow \emptyset$
for each $r_i \in R$ *do*
  if $F_{match}(d, r_i) = 1$ *then*
    Compute priority score:
    $Gpriority(r_i) = \alpha \cdot U(ri) + \beta \cdot W(ri) + \gamma \cdot A(ri)$
    Add r_i *to* $R_{eligible}$
  end if
end for
Select the recipient

$$r^* = \arg \max_{r_i \in R_{\text{eligible}}} G_{\text{priority}}(r_i)$$

Record the transaction $(d, o, r^*, G_{priority}(r^*))$ *on the blockchain ledger.*
Allocate an organ o *to the recipient* r^*

Algorithm 1. Organ Allocation Algorithm.

Organ Allocation Algorithm

Algorithm 1 summarises the complete allocation process by combining cryptographic authentication, compatibility validation, and priority-based recipient selection into a single executable workflow implemented through smart contracts.

QUEUING SYSTEM IMPLEMENTATION

Organ allocation requires precise clinical decision-making, adherence to ethical principles of fairness, and efficient logistical coordination. The proposed solution integrates blockchain technology with a real-time queuing system that utilises clinical data, organ viability constraints, and clearly defined waiting-order protocols. The system performs dynamic recipient ranking based on multiple medical and logistical factors, thereby directing organs to recipients with the highest medical urgency and optimal compatibility. This section presents the tiered priority mechanism, the mathematical formulation of queue scoring, and the integration of blockchain to enhance auditability and trust.

Dynamic Priority Tiers

A three-tier queue is employed to prioritise recipients: Tier 1 (Emergency), such as acute liver failure; Tier 2 (Chronic Deterioration); and Tier 3 (Stable, long-term waitlist). Real-time data updates from IoT sensors and EHRs dynamically adjust recipient status, which is enforced through smart contracts.

Mathematical Representation

A priority scoring function based on complex clinical evaluations defines the rules for queue management, utilising the following variables: the set of available donors $D = \{d_1, d_2, ..., d_n\}$, the set of registered recipients $R = \{r_1, r_2, ..., r_m\}$, the set of viable organs $O = \{o_1, o_2, ..., o_k\}$, and T, the time-stamp of a transaction or queue update. The Organ Viability Check (OVC) is performed using Eq. (3).

$$H_{\text{availability}}(o_i) = \begin{cases} 1, & \text{if viability constraints hold} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

This function ensures that only organs meeting preservation constraints (e.g., temperature and pH limits) are considered for allocation. The queuing layer extends the base priority model defined in the "Proposed Model for Blockchain-Based Organ Transplantation" section by incorporating an explicit compatibility term $C(r_j)$, reflecting real-time allocation constraints. The refined priority

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score is calculated using the Priority Scoring Function defined in Eq. (4).

$$G_{\text{priority}}(r_j) = \alpha \cdot U(r_j) + \beta \cdot W(r_j) + \gamma \cdot A(r_j) + \delta \cdot C(r_j) \quad (4)$$

Where $U(r_j)$ denotes the urgency score (on a scale from 0 to 10, with ICU cases = 10), $W(r_j)$ is the normalised wait time in months, $A(r_j)$ represents the age-based weight (with pediatric cases set to 1.0 and adults to 0.7), and $C(r_j)$ is the compatibility score derived from the matching function $F_{\text{match}}(d_i, r_j)$. The corresponding weight parameters are α = 0.4, β = 0.3, γ = 0.2, and δ = 0.1, which ensure consistency between urgency, fairness, and biological suitability.

While Algorithm 1 defines the core allocation logic at the smart contract level, Algorithm 2 operationalises this logic within the real-time priority queue. The queue assignment algorithm iterates through all registered recipients in descending order of their computed priority scores, assigning the organ to the first recipient who satisfies both compatibility and organ viability criteria, as detailed in Algorithm 2.

This algorithm runs in O($m$log$m$) time due to the sorting step, where m is the number of registered recipients.

Input: $d \in D$, $r_j \in R$, and $o_i \in O$
Output: Allocation of organ o_i to the highest-priority eligible recipient
Sort recipients in R in descending order of priority score
for each recipient $r_j \in R$ do
  if $F_{match}(d, r_j)$=1 and $H_{availability}(o_i)$=1 *then*
    Allocate organ o_i to recipient r_j
    break // highest-priority eligible recipient selected
  end if
end for

Algorithm 2. Queue Assignment Algorithm.

Recipient	*U*	*W*	*A*	*C*	$G_{priority}$
$r1$	10	0.80	0.70	0.90	8.57
$r2$	6	0.50	1.00	0.80	5.98
$r3$	8	0.60	0.90	0.70	7.23
$r4$	4	0.30	0.50	0.60	3.45
$r5$	9	0.70	0.80	0.90	8.12

TABLE 2. Simulated Priority Scores For Organ Allocation.

Case Scenario: Priority Score-Based Allocation

To validate the real-world applicability of our model, we present a case scenario based on hypothetical transplant candidates. Each candidate is evaluated using the defined scoring function $G_{\text{priority}}(r_j)$, incorporating clinical urgency (U), wait time (W), age factor (A), and compatibility (C). As shown in Table 2, recipient r_1 achieves the highest priority score and is selected for organ allocation based on the queue assignment logic. This simulation demonstrates the model's capability to assign organs transparently and systematically. A full end-to-end case narrative is discussed in the "Queuing System Implementation" section.

Blockchain Integration

The queuing system integrates with blockchain to ensure trust, immutability, and auditability of allocation decisions.

- On-Chain Verification: Every match and priority score is cryptographically signed and recorded on-chain using smart contracts, ensuring verifiability.
- Auditability: Regulatory bodies can access the blockchain ledger to audit allocation decisions via digital signatures and immutable logs.

As shown in Fig. 2, the queuing system continuously receives real-time updates from IoT-enabled organ containers and EHR systems, which are synchronised with the blockchain to maintain an up-to-date and tamper-proof queue state.

PERFORMANCE EVALUATION AND RESULTS

Simulation Setup

To evaluate the proposed blockchain-based organ transplantation system, a Hyperledger Fabric (v2.5) simulation was deployed using Dockerised peers and certificate authorities. Synthetic datasets were generated to model donor, recipient, and organ parameters, while three participating organisations executed smart contracts for matching and prioritisation across 1,000 blockchain transactions. As illustrated in Fig. 3, the simulation utilises a layered technology stack that integrates frontend dashboards, backend services, off-chain storage, IoT monitoring, and the Hyperledger Fabric blockchain.

Evaluation Metrics

The performance was evaluated based on the following metrics:

- Matching Speed (ms): Average time taken to identify and match a donor to a recipient using smart contract logic, compared to a traditional manual, centralised allocation system.
- Error Rate Reduction (%): Decrease in data tampering, fraudulent claims, and duplicate entries due to immutable blockchain logging.
- System Throughput (tps): Number of successful organ allocation transactions processed per second.
- Audit Latency (s): Time is taken to verify the legitimacy of a match and trace transaction history.
- Data Integrity Score (%): Calculated based on SHA-256 verification consistency and tamper detection capability.

Quantitative Results

Table 3 compares the performance of the proposed blockchain-based system with a conventional manual allocation approach.

Results Analysis

The blockchain-based implementation reduced the average matching time by over 75% compared to a traditional, centralised database approach.

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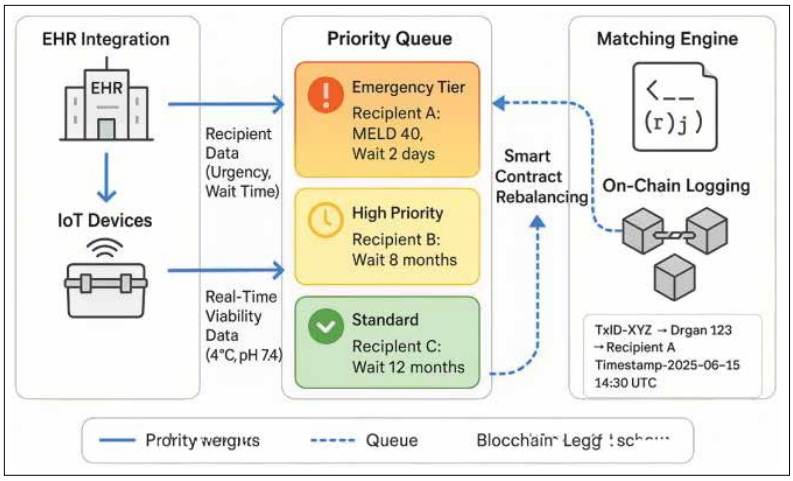


FIGURE 2. Priority Queue Workflow with Real-Time IoT and EHR Updates.

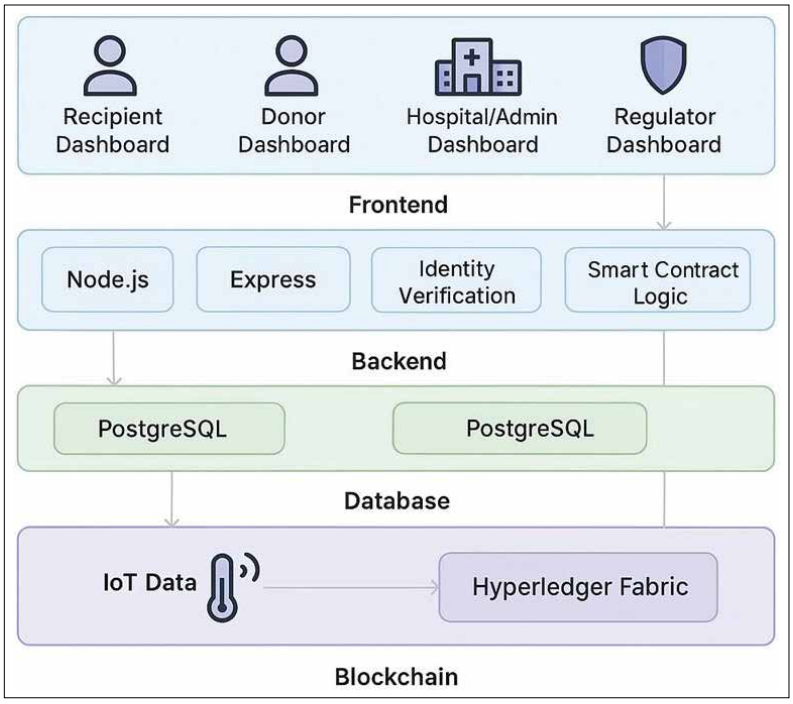


FIGURE 3. Technology Stack Diagram for Blockchain-Based Organ Transplant Management System.

Metric	Blockchain System	Manual System
Matching Speed (ms)	210	980
Error Rate Reduction (%)	92.5	61.3
System Throughput (tps)	27.4	4.6
Audit Latency (s)	0.45	3.2
Data Integrity Score	99.9%	82.4%

TABLE 3. Performance Comparison With Baseline System.

Specifically, matching speed improved from 980ms (manual system) to 210 ms (blockchain system). Furthermore, the integration of SHA-256 hashing, distributed peers, and smart contracts enabled near real-time auditability, contributing to a 92.5% reduction in administrative errors, duplicate entries, and fraudulent modifications. System throughput increased nearly sixfold due to automated rule execution and decentralised consensus.

Scalability and Robustness

Stress testing with increasing transaction loads (from 100 to 10,000 events) revealed linear scaling in performance, accompanied by minimal latency spikes, which affirmed the framework's scalability and robustness under peak demand scenarios, such as mass casualty events. The evaluation confirms that blockchain-enabled automation in organ allocation enhances operational efficiency, improves fairness through auditable smart contracts, and upholds transparency through immutable records. These features collectively contribute to more ethical and timely organ transplantation workflows.

Illustrative Case Study

To validate the real-world applicability of the proposed blockchain framework, we present a comprehensive case study that demonstrates the logical execution flow of the system rather than a statistically representative deployment, covering the organ transplant process from donor registration to organ delivery.

- Step 1 – Donor Registration: A donor (Donor ID: D1001) registers and provides digital consent using a blockchain interface. Their organ data, such as blood type, HLA profile, and medical history, is securely hashed and stored on-chain.
- Step 2 – Recipient Entry: A recipient (Recipient ID: R2003) is registered in the system. Clinical urgency, biological compatibility, and waiting time parameters are updated based on linked EHR inputs and system records.
- Step 3 – Matching and Prioritisation: The system has an inbuilt smart contract that executes the matching function F_{match} (D1001, R2003) to verify compatibility. Subsequently, the priority function Gpriority(R2003) is used to calculate a score based on urgency (0.10), wait time (0.75), age (0.85), and compatibility (0.95), resulting in the top priority within the simulated queue.
- Step 4 – Allocation and Tracking: Based on the computed priority order, the organ is allocated to recipient R2003. The IoT sensors embedded in the transport container continuously monitor temperature, humidity, and geolocation, storing all transit data immutably on-chain.
- Step 5 – Delivery Confirmation: The blockchain records the final delivery status as the organ arrives at the hospital of the recipient. All steps in the process are verifiable and tamper-proof, which ensures transparency and ethical compliance.

This case study serves as a representative example illustrating how the proposed framework operates at the individual transaction level. System-level performance, scalability, and fairness are evaluated separately through large-scale simulations in the "Performance Evaluation and

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Results" section, thereby avoiding overgeneralisation from a single-case instance.

Discussion on Limitations and Trade-Offs

Despite the many advantages of the proposed framework, several limitations and risks remain. The framework assumes reliable real-time IoT data and uninterrupted blockchain infrastructure, which may be challenging in rural or low-resource healthcare settings due to limited connectivity, constrained computational resources, and sparse IoT deployment. Data privacy breaches also remain a concern, particularly in cases of misconfigured access control policies or compromised key management mechanisms. While smart contracts automate allocation and improve efficiency, they may lack the flexibility to accommodate unforeseen clinical emergencies or urgent policy overrides. Additionally, interoperability with existing hospital EHR systems and regulatory compliance across different jurisdictions pose non-trivial challenges. The permissioned nature of Hyperledger Fabric further necessitates robust infrastructure and governance models. Addressing these trade-offs is essential for achieving scalable, secure, and large-scale real-world adoption.

Conclusion and Future Directions

This study presents a blockchain-based framework for organ transplantation using Hyperledger Fabric, smart contracts, and cryptographic mechanisms (SHA-256, SSI) to enhance transparency, efficiency, and trust. The system's immutable records and automated matching algorithms outperform traditional methods in speed and accuracy while ensuring ethical compliance. However, scalability challenges in large-scale deployments and regulatory harmonisation across regions remain key hurdles. Future work should explore AI-driven predictive matching, hybrid blockchain models for integrating legacy systems, and federated networks for cross-border organ sharing. Off-chain computation methods could address scalability limitations. By overcoming these challenges, this framework can transform organ transplantation into a more secure, efficient, and globally accessible healthcare solution.

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